

Data catalog components

So far, you've learned that data catalogs are centralized inventories of an organization's data assets. These inventories provide a comprehensive view of available data so that users can access, understand, and use data assets more effectively. Maintaining an accurate catalog is foundational for effective data management. You also learned about the components of a data catalog, and how they function in an organization. In this reading, you'll explore these components in more detail.

Components of data catalogs

Data catalogs can include a variety of components outlined in your metadata, like structured data, unstructured data, reports, dashboards and visualizations, and any automations and machine learning models you've created. Review the chart to learn more about these data components:

Component	Definition	Examples
Structured data	Data organized in a certain format, like rows and columns	• Expense reports • Tax returns • Retail inventory • Customer information • Statistics and numerical data
Unstructured data	Data that's not organized in any easily identifiable way	• Social media posts • Emails • Videos • Images • Books • Product reviews
Reports	A static collection of data periodically given to stakeholders	• Payroll • Employee performance • Sales reports • Quarterly revenue
Dashboards and visualizations	A comprehensive display that utilizes data visualizations and visualization tools to present information	• Dashboard monitoring • Live social media engagement

		• Charts and graphs communicating insights to stakeholders
Automations and machine learning models	Automatic processes, and the development of algorithms and statistical models to teach computer systems to analyze and discover patterns in data	• Linear regression model • Decision tree • Pipelines

Key takeaways

Data catalogs are key tools for several reasons. For example, they help you manage your metadata, which contains important information about your data. They also help you track the data lineage across systems, so you can identify any potential issues and have confidence in its integrity. Finally, data catalogs facilitate data governance by establishing policies, controls, and guidelines for data use, security, and privacy. All of these elements ensure the quality of the data you're working with, so you can be confident in your analysis.